

Class 8: Unsupervised Learning Mini-Project

Hubert Phan (PID: A17663330)

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Exploratory data analysis

Preparing the data

```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)

# Examine df
```

```
head(wisc.df, 4)

# Remove diagnosis column for clustering
wisc.df <- wisc.df[ , -1]

# Create diagnosis vector for later
diagnosis <- as.factor(read.csv(fna.data)$diagnosis)
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.df)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
sum(diagnosis == "M")
```

```
[1] 212
```

Q3. How many variables/features in the data are suffixed with `__mean`?

```
# Check which colname contains "__mean"
mean_cols <- grep("__mean$", colnames(wisc.df))
length(mean_cols)
```

```
[1] 10
```

Principal Component Analysis

Check the mean and standard deviation of the features (i.e. columns) of the `wisc.data` to determine if the data should be scaled.

```
# hide results, show CV calculation only
# Check column means and standard deviations
colMeans(wisc.df)
```

```
#STANDARD DEVIATION
apply(wisc.df, 2, sd)
```

```
#consider applying CV rule to determine if scaling is necessary
#say that the rule is that if most of CV > 0.3, then scaling is necessary
cv <- apply(wisc.df, 2, sd) / colMeans(wisc.df)
cv
```

radius_mean	texture_mean	perimeter_mean
0.2494497	0.2229712	0.2642083
area_mean	smoothness_mean	compactness_mean
0.5373645	0.1459536	0.5061555
concavity_mean	concave.points_mean	symmetry_mean
0.8977525	0.7932036	0.1513248
fractal_dimension_mean	radius_se	texture_se
0.1124304	0.6844320	0.4533400
perimeter_se	area_se	smoothness_se
0.7054476	1.1277714	0.4264347
compactness_se	concavity_se	concave.points_se
0.7028841	0.9464579	0.5230768
symmetry_se	fractal_dimension_se	radius_worst
0.4024073	0.6972696	0.2970794
texture_worst	perimeter_worst	area_worst
0.2393661	0.3132777	0.6465681
smoothness_worst	compactness_worst	concavity_worst
0.1724913	0.6187893	0.7664699
concave.points_worst	symmetry_worst	fractal_dimension_worst
0.5735495	0.2132805	0.2151539

```
#look at summary statistics of CV
summary(cv)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1124	0.2419	0.4797	0.4869	0.6941	1.1278

Based on the results above, the data should be scaled before performing PCA.

```
# perform PCA on the wisc.df data frame, scaling the variables
wisc.pr <- prcomp( wisc.df, center = TRUE, scale. = TRUE)

# Look at summary of results
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

Ans: Proportion of variance: 0.4427

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

Ans: At least 3 PCs are required to describe at least 70% of the original variance in the data.

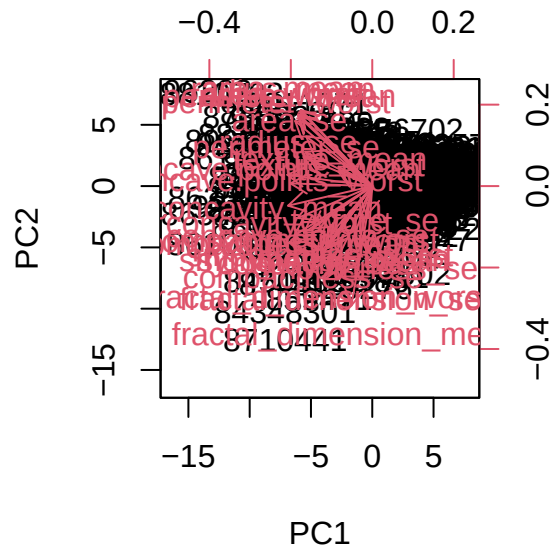
Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

Ans: At least 7 PCs are required to describe at least 90% of the original variance in the data.

Interpreting PCA results

Create a biplot of the wisc.pr using the biplot() function.

```
# Create biplot of PCA results
biplot(wisc.pr, scale = 0)
```



Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

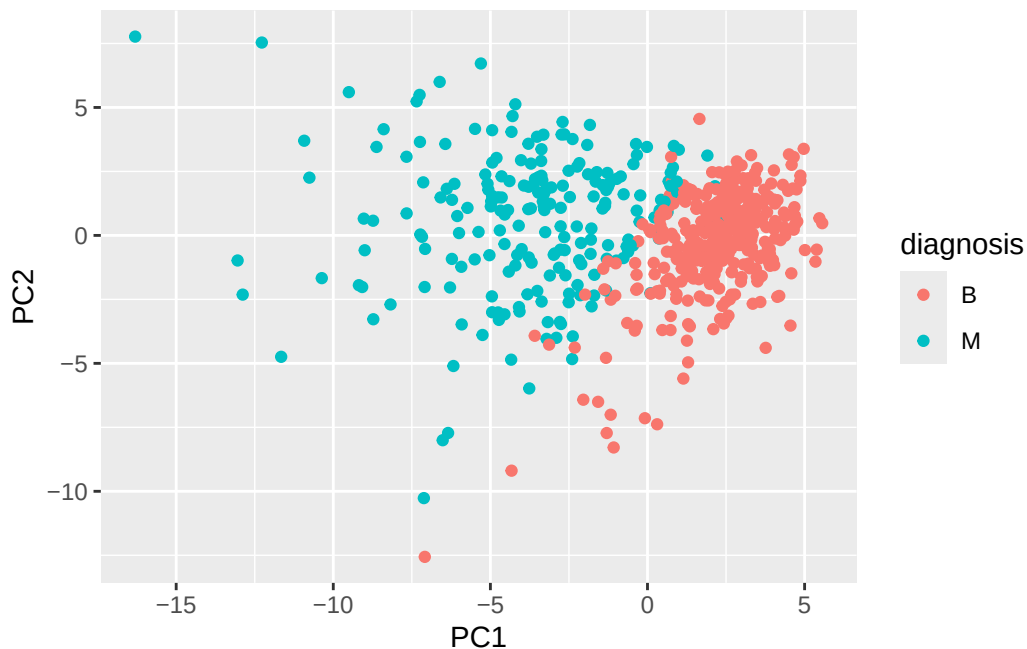
Ans: What stands out is that the plot is quite cluttered with many points, making it difficult to interpret. I am unsure of where the points lie and which labels correspond to which points.

Lets generate a more standard scatter plot of each observation along principal components 1 and 2 (i.e. a plot of PC1 vs PC2 available as the first two columns of `wisc.pr$x`) and color the points by the diagnosis (available in the diagnosis vector you created earlier).

```
# Scatter plot observations by components 1 and 2
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.2

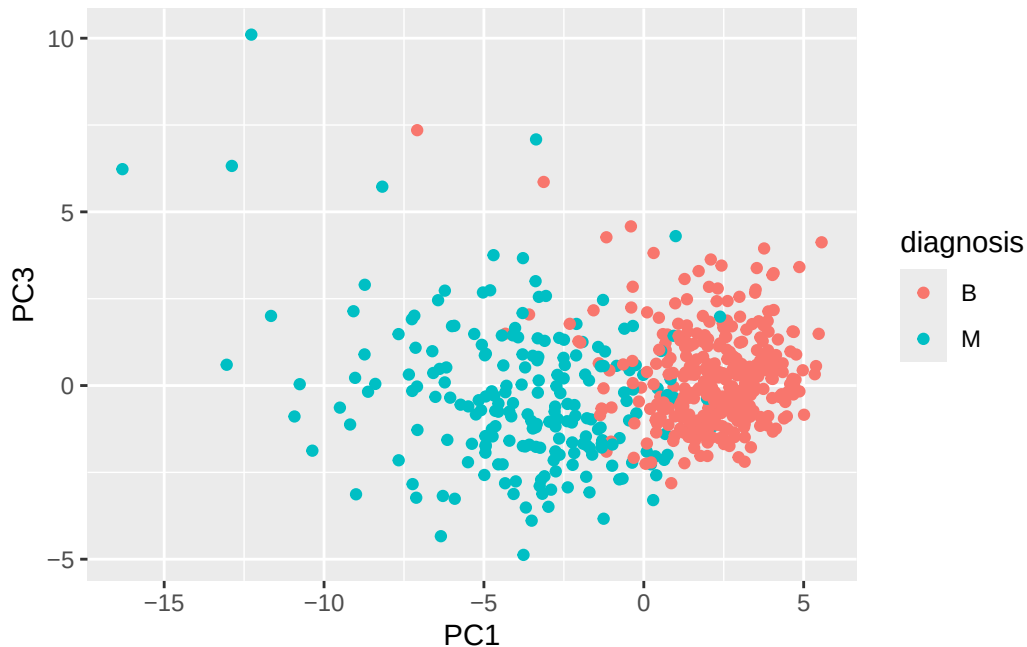
```
ggplot(wisc.pr$x) +
  aes(x = PC1, y = PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
# Scatter plot observations by components 1 and 3
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(x = PC1, y = PC3, col=diagnosis) +
  geom_point()
```



Ans: There is greater variability among malignant samples along PC3 compared to PC2. PC3 primarily captures variability within diagnosis groups rather than separation between benign and malignant samples. While PC1 shows clear separation between diagnoses, the increased spread observed in PC3 reflects greater within-class heterogeneity among malignant samples.

Variance explained

Calculate the variance of each principal component by squaring the sdev component of wisc.pr (i.e. `wisc.pr$sdev^2`). Save the result as an object called `pr.var`.

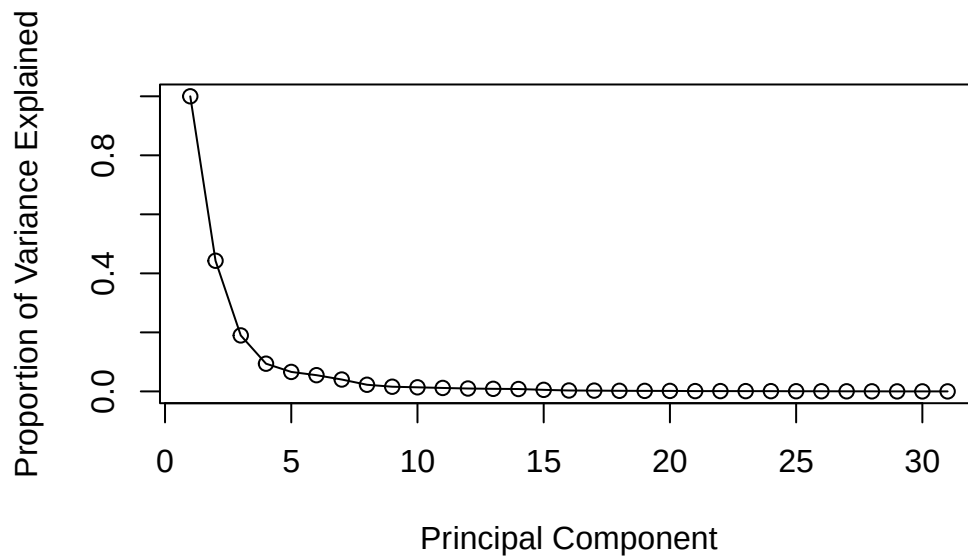
```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

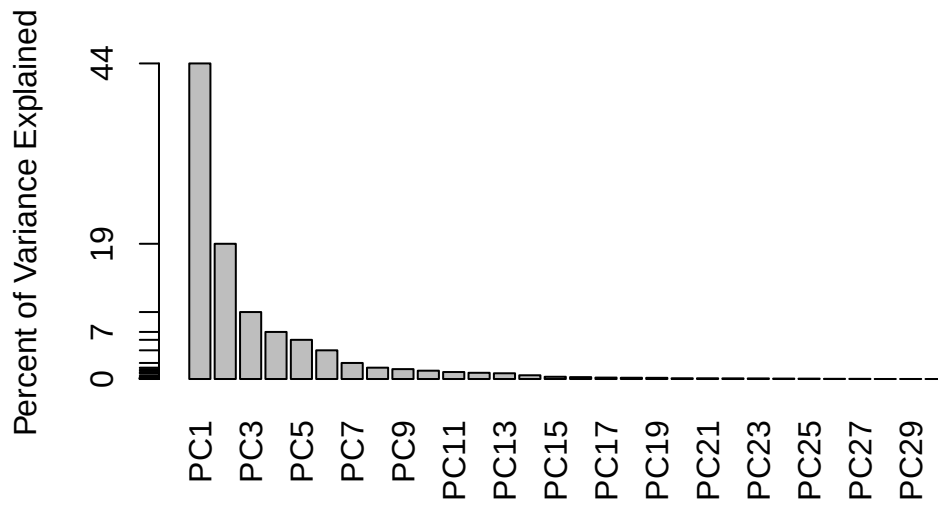
Calculate the variance explained by each principal component by dividing by the total variance explained of all principal components. Assign this to a variable called `pve` and create a plot of variance explained for each principal component.

```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Communicating PCA results

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
# Obtain concave.points_mean for PC1 loading vector
wisc.pr$rotation["concave.points_mean",1]
```

```
[1] -0.2608538
```

```
# look at largest magnitude point_mean among all PC
# Absolute loadings for concave.points_mean across all PCs
abs_loadings <- abs(wisc.pr$rotation["concave.points_mean", ])

# Find PC with largest absolute contribution
pc_max <- colnames(wisc.pr$rotation)[which.max(abs_loadings)]

# Obtain max
max_loading <- abs_loadings[pc_max]
```

```
pc_max
```

```
[1] "PC27"
```

```
max_loading
```

```
      PC27  
0.4546994
```

Ans: Although `concave.points_mean` has a large loading magnitude on PC27, PC27 still has very little overall variance. Therefore, PC27 is biologically less important than PC1, which has substantially more variance and drives separation between benign and malignant samples.

Hierarchical clustering

First scale the `wisc.data` data and assign the result to `data.scaled`.

```
# Scale the wisc.data data using the "scale()" function  
data.scaled <- scale(wisc.df)
```

Next, calculate the (Euclidean) distances between all pairs of observations in the new scaled dataset and assign the result to `data.dist`.

```
data.dist <- dist(data.scaled, method = "euclidean")
```

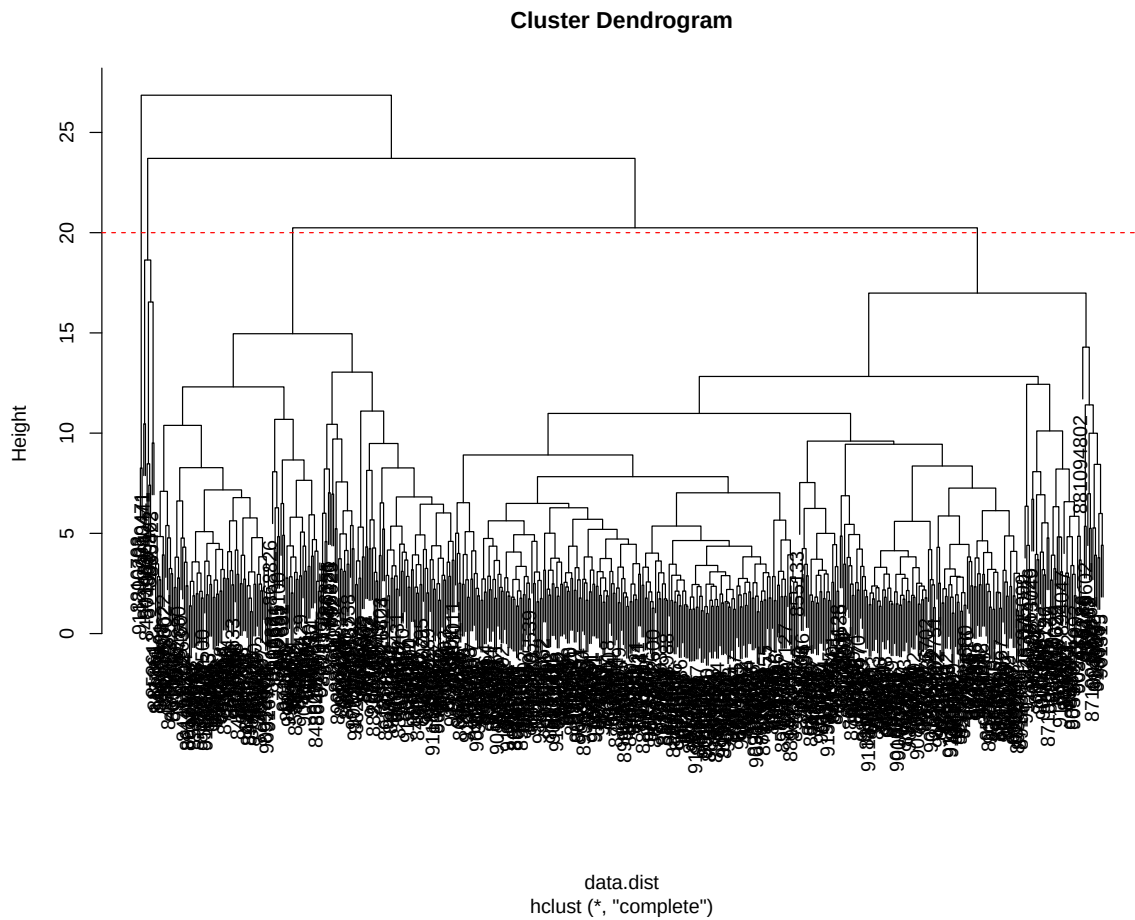
Create a hierarchical clustering model using complete linkage. Manually specify the `method` argument to `hclust()` and assign the results to `wisc.hclust`.

```
wisc.hclust <- hclust(data.dist, method = "complete")
```

Results of hierarchical clustering

Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust)
abline(h = 20, col="red", lty=2)
```



```
# Confirm 4 groups with cutree
table(cutree(wisc.hclust, h = 20))
```

```
1  2  3  4
177 7 383 2
```

Selecting number of clusters

Use `cutree()` to cut the tree so that it has 4 clusters. Assign the output to the variable `wisc.hclust.clusters`.

```
wisc.hclust.clusters <- cutree(wisc.hclust, h = 20)
```

We can use the `table()` function to compare the cluster membership to the actual diagnoses.

```
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Q11. OPTIONAL: Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 6? How do you judge the quality of your result in each case?

```
# Cut into 2 clusters
wisc.hclust.clusters2 <- cutree(wisc.hclust, k = 2)
table(wisc.hclust.clusters2, diagnosis)
```

	diagnosis	
wisc.hclust.clusters2	B	M
1	357	210
2	0	2

```
# Cut into 6 clusters
wisc.hclust.clusters6 <- cutree(wisc.hclust, k = 6)
table(wisc.hclust.clusters6, diagnosis)
```

	diagnosis	
wisc.hclust.clusters6	B	M
1	12	165
2	0	5
3	331	39
4	2	0
5	12	1
6	0	2

Ans: Based on the known diagnosis classes, clustering quality should be judged by comparing cluster assignments to known labels, with better results showing clusters dominated by a certain diagnosis. Furthermore, increasing the number of clusters should be examined carefully as over segmentation sometimes doesn't represent true diagnostic separation.

Using different methods

Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

Ans: For the data.dist dataset, Ward.D2 is optimal because it forms compact, low variance clusters that favor the diagnosis class-like structure of the data. Ward.D2 also produces clusters with higher purity and clearer interpretability than the other methods which either over-chain points or do not explicitly optimize variance.

Combining methods

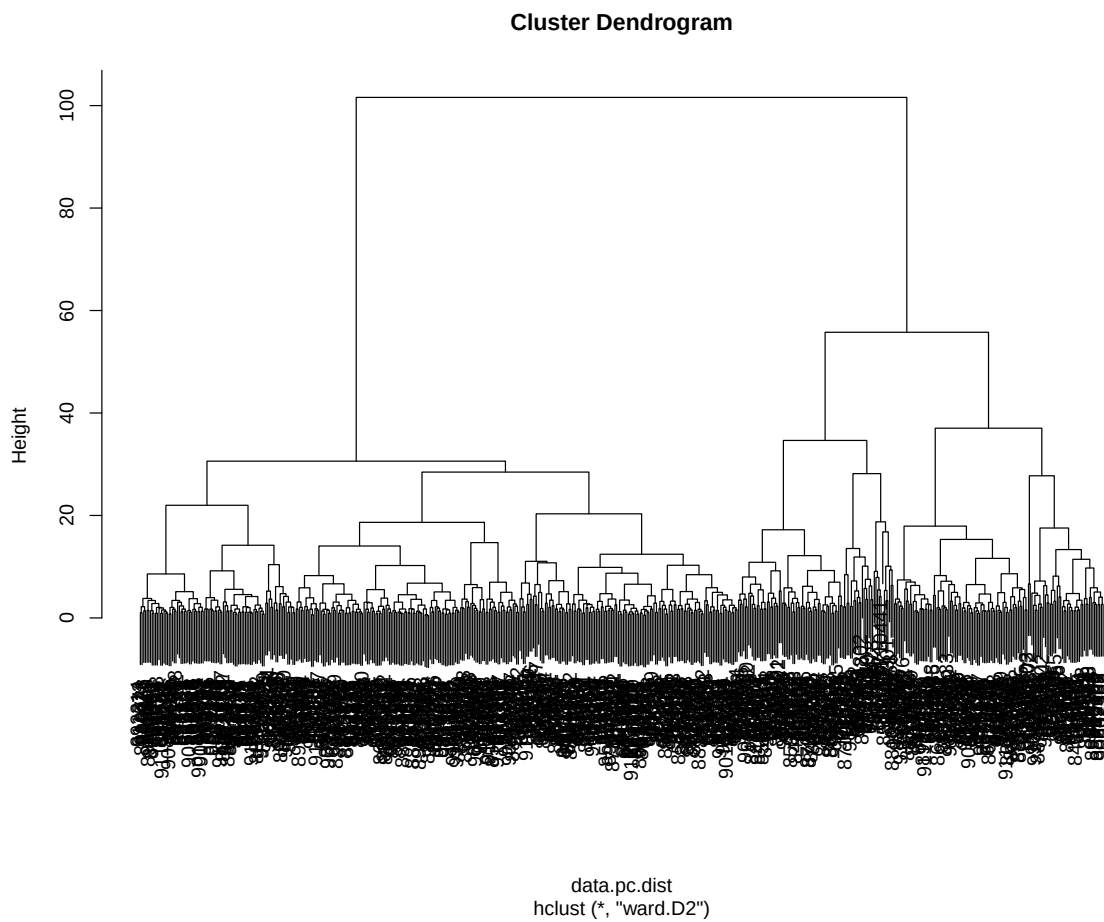
Clustering on PCA results

So far we have tried PCA and hierarchical clustering separately. Now let's combine them: cluster on the PC scores instead of the original 30 features.

Using the minimum number of principal components required to describe at least 90% of the variability in the data, create a hierarchical clustering model with the linkage method="ward.D2". We use Ward's criterion here because it is based on multidimensional variance like principal components analysis. Assign the results to wisc.pr.hclust.

```
# Create data.pc.dist. Only consider the first 7 PC
data.pc <- wisc.pr$x[, 1:7] # take first 7 PC scores
data.pc.dist <- dist(data.pc) # Euclidean distance on PC scores
# Hierarchical Clustering with ward.D2
wisc.pr.hclust <- hclust(data.pc.dist, method = "ward.D2")
```

```
plot(wisc.pr.hclust)
```



Group data, compare cluster assignments to labels, and re-examine PC1 vs. PC2 with new grouping

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
table(wisc.pr.hclust.clusters)
```

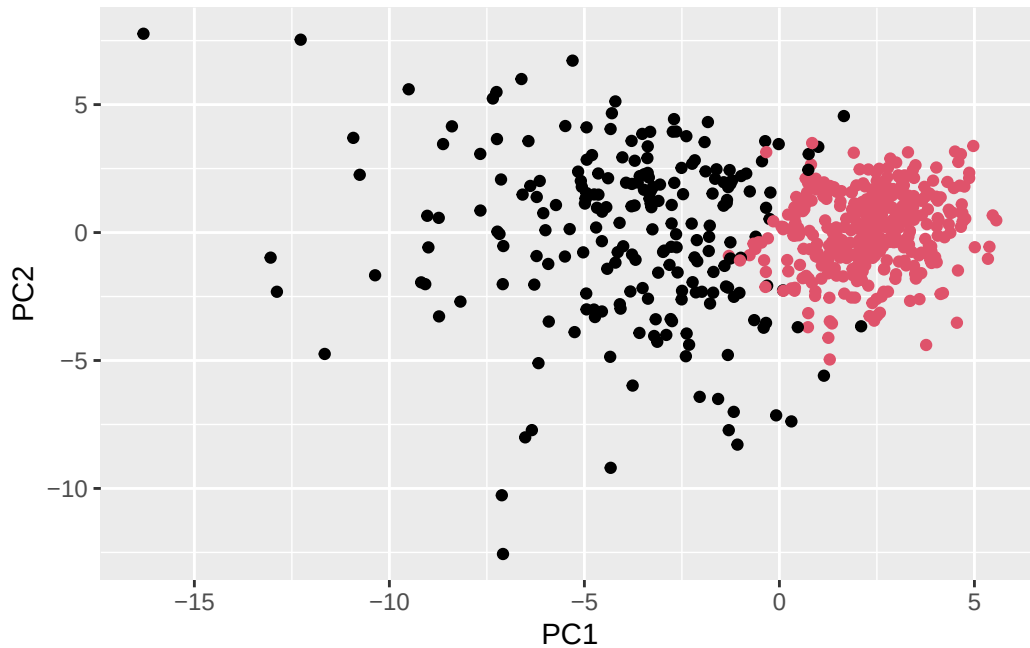
```
wisc.pr.hclust.clusters
  1  2
216 353
```

```
table(wisc.pr.hclust.clusters , diagnosis)
```

```
              diagnosis
wisc.pr.hclust.clusters  B  M
```

1	28	188
2	329	24

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=wisc.pr.hclust.clusters)
```



Q13. How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?

```
# Compare to actual diagnoses
table(wisc.pr.hclust.clusters , diagnosis)
```

	diagnosis	
wisc.pr.hclust.clusters	B	M
1	28	188
2	329	24

Ans: The newly created hclust model produces clusters that are either dominated by “M” or “B” diagnoses. This indicates improved alignment with diagnosis class structure.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

```
# compare wisc.hclust.clusters cluster with label
table(wisc.hclust.clusters, diagnosis)
```

```

      diagnosis
wisc.hclust.clusters  B  M
1      12 165
2       2   5
3     343  40
4       0   2

```

```
# compare wisc.pr.hclust.clusters clusters with label
table(wisc.pr.hclust.clusters, diagnosis)
```

```

      diagnosis
wisc.pr.hclust.clusters  B  M
1      28 188
2     329  24

```

Ans: The hierarchical clustering on the original features shows greater mixing of diagnosis across clusters, indicating weaker separation by diagnosis. The clustering on the PCA specific data produces clusters that are dominated by a single diagnosis, indicating better separation after applying PCA. We can conclude the PCA cleans our data which allows the clustering to better resemble biological structure.

Sensitivity/Specificity

Q15. OPTIONAL: Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

```
#create sensitivity and specificity functions
#create function to determine majority cluster label
#Find TP,FN,TN,FP
#compare PCA approach vs original features

sensitivity <- function(TP, FN) {
```

```

    if ((TP + FN) == 0) return(NA_real_)
    TP / (TP + FN)
}

specificity <- function(TN, FP) {
  if ((TN + FP) == 0) return(NA_real_)
  TN / (TN + FP)
}

# Function to assign cluster labels by majority class
assign_cluster_labels <- function(clusters, true_labels) {
  # Create contingency table
  tab <- table(clusters, true_labels)

  # Assign each cluster the label with the highest count
  cluster_labels <- apply(tab, 1, function(x) {
    names(which.max(x))
  })

  return(cluster_labels)
}

evaluate_clustering <- function(clusters, true_labels) {
  # Assign cluster labels by majority vote
  cluster_labels <- assign_cluster_labels(clusters, true_labels)

  # Predicted label for each observation
  predicted_labels <- cluster_labels[as.character(clusters)]

  # Confusion matrix
  conf_mat <- table(Predicted = predicted_labels, Actual = true_labels)

  # Extract confusion matrix components
  TP <- conf_mat["M", "M"]
  FN <- conf_mat["B", "M"]
  TN <- conf_mat["B", "B"]
  FP <- conf_mat["M", "B"]

  # Compute metrics
  sens <- sensitivity(TP, FN)
  spec <- specificity(TN, FP)

  # Return results

```

```
list(
  confusion_matrix = conf_mat,
  sensitivity = sens,
  specificity = spec
)
```

```
# Do analysis on pca approach
results_pca <- evaluate_clustering(
  clusters = wisc.pr.hclust.clusters,
  true_labels = diagnosis
)

results_pca$confusion_matrix
```

	Actual	
Predicted	B	M
B	329	24
M	28	188

```
results_pca$sensitivity
```

```
[1] 0.8867925
```

```
results_pca$specificity
```

```
[1] 0.9215686
```

```
#do analysis on original features
results_original <- evaluate_clustering(
  clusters = wisc.hclust.clusters,
  true_labels = diagnosis
)

results_original$confusion_matrix
```

	Actual	
Predicted	B	M
B	343	40
M	14	172

```
results_original$sensitivity
```

```
[1] 0.8113208
```

```
results_original$specificity
```

```
[1] 0.9607843
```

Ans: Hierarchical clustering on PCA scores improved sensitivity (0.8867925 vs 0.8113208), indicating better grouping of malignant samples, while clustering on the original features showed slightly higher specificity (0.9607843 vs 0.9215686).

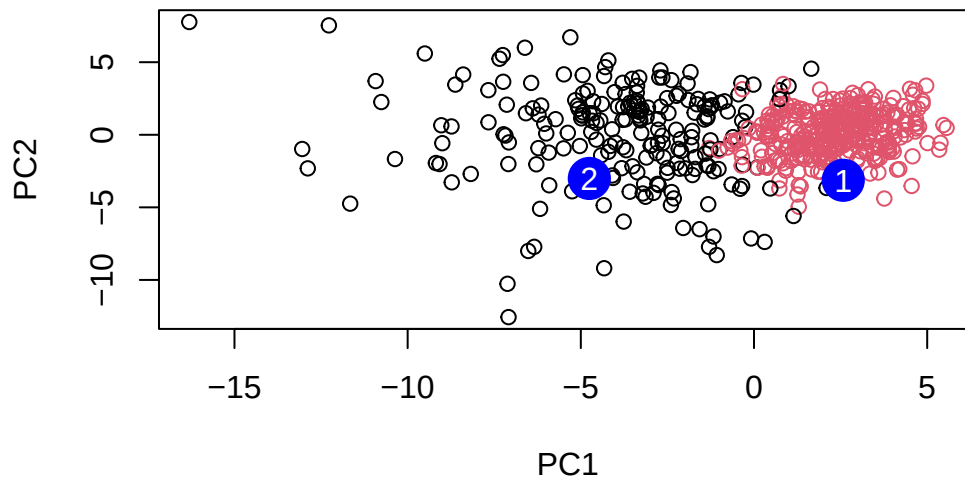
Prediction

We will use the `predict()` function that will take our PCA model from before and new cancer cell data and project that data onto our PCA space.

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=wisc.pr.hclust.clusters)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Ans: Patient 2 should be prioritized as their PCA projection aligns with malignant samples while patient 1 aligns with benign samples.